Unit-1 :-

→ Set Theory, Functions & Counting
(Hours - 6) (% 12%)

* Set Theory :

Basic Concepts of set Theory :-
Definitions, Inclusion, Equality of sets, Cartesian Product, The power set, Some operations on Sets, Venn Diagrams, Some basic Set identities.

→ Defn (Set) : A well defined ~~set~~ Collection of objects is called a set.

Ex-: 1 ① Family is a set of persons
② Class is a set of students.
③ N, Z, Q, R, C are the sets of natural, integers, rational, real and Complex numbers.

Methods of Discribing Sets

Roster Method
(Listing Method)

Ex $N = \{1, 2, 3, 4, 5, \ldots\}$

Set Builder form
(Property Method)

Ex $A = \{ x ; x^2 - 9 = 0, x \in R \}$

→ Defn (Subset) :-
Let A and B be two sets. If each element of set A is an element of set B then A is called a subset of B, and denoted by $A \subset B$.

Ex :- 2: $A = \{1, 2, 3\}$ $B = \{1, 2, ..., 10\}$ then $A \subset B$ but $B \not\subset A$.

→ Defn (Proper subset) :
If A is a subset of B and no. of elements of A & B are not same then A is called proper subset of B.

Ex : 3 In ex-2 A is called proper subset of B

→ Defn (Equality of sets) :
Two sets A & B are called equal sets if $A \subset B$ and $B \subset A$.

→ Defn (Empty / Null set)
The set having no elements is called Empty set.

→ Defn (Universal set)
A non-empty set of which all the sets under consideration are subsets is called Universal set.

→ Ordered pair (a, b) indicates $a \in A$ & $b \in B$ for two sets A and B.

→ Defn : (Cartesian Product)
Let A and B be two sets then $A \times B$ is called cartesian Product of A & B.

Ex-4 : Let $A = \{-1, 0, 1\}$ $B = \{p, q\}$

then $A \times B = \{(-1, p), (-1, q), (0, p), (0, q), (1, p), (1, q)\}$

Rmk : (1) If A contains m elements & B contains n elements then $A \times B$ contains $m \times n$ elements.

→ Defn : (Power set)
If S is any set ~~and~~ then set of all subsets of S is called power set of S and denoted by $P(S)$.

Ex-4 : Let $S = \{1, 2, 3\}$ then

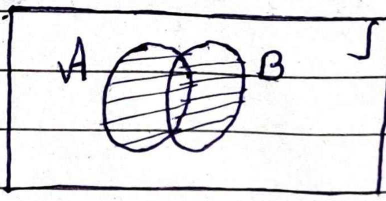
$P(S) = \{\phi, \{1\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{2\}, \{3\}, S\}$

NOTE : If a set S has n element, then its power set has 2^n elements.

☆ Some operations on sets :

(1) Union :-

"For any two sets A and B, the union of A and B is set of all those elements which are either in A or in B or in both sets, & denoted by $A \cup B$."



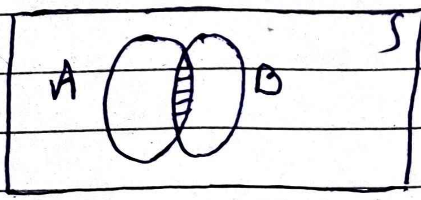
$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Ex-5 :- Let $A = \{-2, -1, 0, 1, 2\}$ and $B = \{-1, 0, 1, 2, 3, 4, 5\}$

then $A \cup B = \{-2, -1, 0, 1, 2, 3, 4, 5\}$

(2) Intersection :-

"For any two sets A and B, the intersection of A and B is the set of all those elements which are present in A & also in B, and denoted by $A \cap B$"



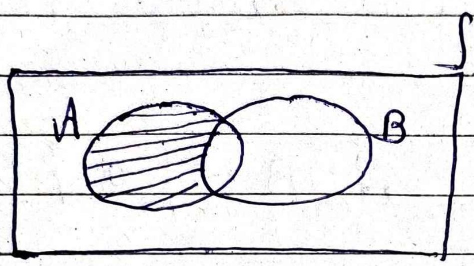
$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Ex-6 :- In Ex-5, $A \cap B = \{-1, 0, 1, 2\}$.

NOTE: If $A \cap B = \phi$, then both are called disjoint set.

(3) Difference of Two Sets :-

"Four any two sets A and B, the difference of A & B is denoted by A - B is the set of all elements which are present in A but not in B."



$A - B = \{x \mid x \in A \ \& \ x \notin B\}$

Ex-7 :- Let $A = \{1, 2, \ldots, 10\}$ & $B = \{5, 6, \ldots, 14\}$

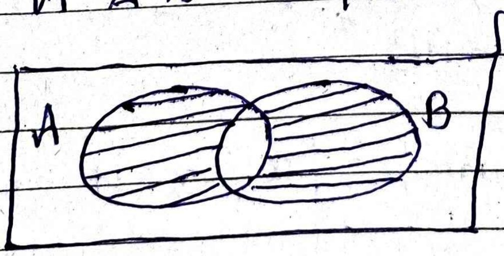
Then $A - B = \{1, 2, 3, 4\}$ &

$B - A = \{11, \ldots, 14\}$

(4) Symmeteric difference :

Four any two sets A and B, sym. diff. is denoted by $A \oplus B$ or $A \Delta B$ &

$A \Delta B = \{A \cup B\} - \{A \cap B\}$



Ex-8 :- In Ex-7 $A \cup B = \{1, 2, \ldots, 14\}$ &

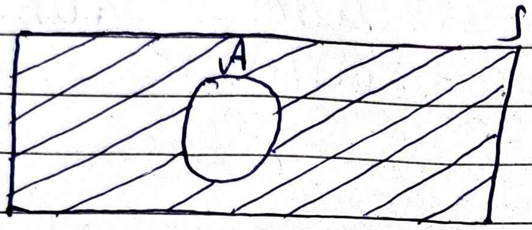
$A \cap B = \{5, 6, \ldots, 10\}$ then

$A \Delta B = (A \cup B) - (A \cap B)$

$= \{1, 2, 3, 4, 11, 12, 13, 14\}$

(5) Complement of a Set

For any set A, Complement of A is denoted by A' or A^c is the set of all elements which are in universal set S but not in A.



$A' = \{x \mid x \notin A \ \& \ x \in S\}$

Eg: Let $S = \{1, 2, ..., 10\}$ & $A = \{1, 2, ..., 5\}$
then $A' = \{6, 7, ..., 10\}$

Venn Diagrams

[Use Venn diagrams to prove the following]

(1) Let A, B, C be sets such that $A \subseteq B$, $B \subseteq C$, $(B \cap C) \subseteq A$ & $A \subseteq (B \cap C)$
[Just draw diagrams for above cases]

(2) $A \cup (\bar{B} \cap C) = (A \cup \bar{B}) \cap (A \cup C)$

(3) $A \Delta (B \Delta C) = (A \Delta B) \Delta C$

(4) $A \cap B \cap C = A - [(A - B) \cup (A - C)]$

[Above 4 pbms are left for the students to solve]

★ Some basic Set identities :-

(1) Idempotent Laws : $A \cup A = A$, $A \cap A = A$

(2) Commutative Laws : $A \cup B = B \cup A$,
$A \cap B = B \cap A$

(3) Associative Laws : $A \cup (B \cup C) = (A \cup B) \cup C$
& $A \cap (B \cap C) = (A \cap B) \cap C$

(4) Distributive Laws :
(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(5) De Morgan's Laws :
(i) $(A \cup B)' = A' \cap B'$ &
(ii) $(A \cap B)' = A' \cup B'$

(6) Absorption Laws :
(i) $A \cup (A \cap B) = A$ & (ii) $A \cap (A \cup B) = A$

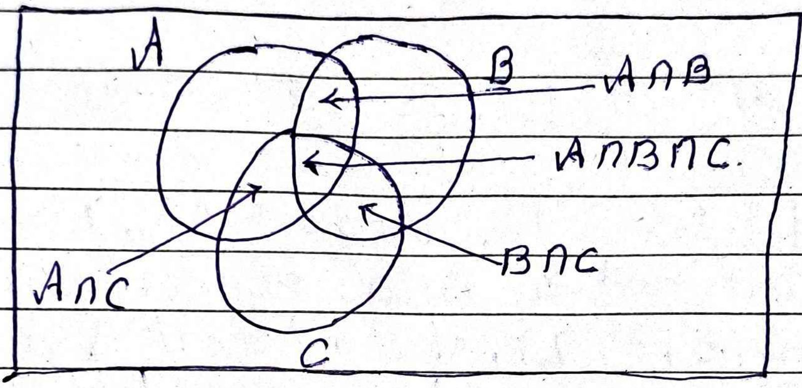
(7) Double Complement : $(A')' = A$

NOTE : Cardinality of A is denoted by $n(A)$ or $|A|$.

Inclusion

(i) $|A \cup B| = |A| + |B| - |A \cap B|$

(ii) $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C|$
$- |C \cap A| + |A \cap B \cap C|$



(iii) $|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D|$

$- [\,|A \cap B| + |A \cap C| + |A \cap D| + |B \cap C|$
$+ |B \cap D| + |C \cap D|\,]$

$+ [\,|A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| +$
$|B \cap C \cap D|\,]$

$- |A \cap B \cap C \cap D|$

Examples based on Set theory :-

Ex-1 :- Let $R = \{x / x^2 = 7\}$, $S = \{x / x + 4 = 4\}$ & $T = \{x / x^2 + 2 = 8\}$. Find out which of them are equal?
[Ans : R & T equal but not S]

Ex-2 : Determine the power set of A for given $A = \{1, 2, 3, 4\}$.

[Ans : $P(A) = \{\phi, \{1\}, ----, \{1,2,3,4\}$ (total 2^4)]

Ex-3 Let $U = \{1, 2, ..., 9\}$, $A = \{1, 4, 9\}$, $D = \{2, 3, 5, 7\}$
$P = \{x / x \in U$ and x is a perfect square$\}$
$R = \{1, 2, 3, 5, 7, 9\}$, $N = \{x / x \in U$ & x is prime$\}$, ϕ
Then (i) Determine which sets are the subsets of others
(ii) Determine which sets are proper subset of others.
(iii) Determine pair of sets which are disjoint

[A : (i) All the sets are subset of U,
ϕ is s.s. of all sets, $A \subset P$, $D \subset R$, $D \subset N$
(ii) All sets are P.S.S. of U, D pr. $\subset R$,
N pr. $\subset R$, ϕ is pr. s.s. of all
(iii) The pairs A & D, A & N, P & D and
P & N are disjoint sets]

Ex-4 Draw the Venn Diagram for
(i) $P \cap Q = P$ (ii) $A \cup (B \cap C)$ (iii) $(A \cap B) \cup (A \cap C)$

Ex-5: A survey of 550 television watchers produced the following information.
285 watch football games;
195 watch hockey games;
115 watch basketball games;
45 watch Football & basket ball;
70 " Football & hockey;
50 " Hockey & basket ball.
100 do not watch any of three games.
then (i) How many people in the survey watch all three games?
(ii) How many people watch exactly one of the three games?

[A :- 20 & 325]

Ex-6 : Among the first 500 positive integers
(i) Determine the integers which are not divisible by 2, nor by 3, nor by 5
(ii) Determine the integers which are exactly divisible by one of them.
[A :- 134 & 232]

Functions :-

Intro. & defn., Codomain, range, image, Value of a function ; Examples, Surjective, injective, bijective ; examples ; Composition of functions, examples ; Inverse function, identity map, Condition of a function to be invertible, examples ; Inverse of composite functions, Properties of Composite function.

★ **Definitions :-**

(1) Function : Let A and B be two non-empty sets. Then a function from A to B is a rule which assigns to each element ~~of~~ $a \in A$ to unique element $b \in B$.

(2) Domain & Co-domain :
Let $f : A \to B$ be a function, then A is called domain & B is called Co-domain of f.

(3) Range :
Range of a function f is

$$R_f = \{ b ; b \in B \ \& \ f(a) = b \text{ for some } a \in A \}$$

Examples :-

[1] Check if the following are functions from A to B or not. If yes find the range also.

(a) $A = \{1, 2, 3, 4\}$, $B = \{b_1, b_2, b_3\}$ &
$f = \{(1, b_1), (2, b_2), (3, b_1)\}$

(b) $A = \{x, y, z, t\}$, $B = \{1, 2, 3, 4, 5, 6\}$
$f = \{(x, 1), (y, 1), (z, 3), (t, 4)\}$

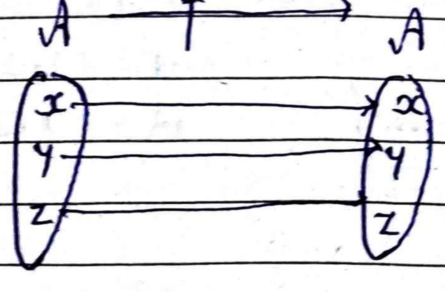
(c) $A = \{x, y, z, t\}$, $B = \{1, 2, 3, 4, 5\}$
$f = \{(x, 1), (t, 4), (z, 2), (y, 3), (t, 5)\}$

[Ans → No, Yes, No]

★ Types of Functions :-

(1) Identity Function(Mapping) :

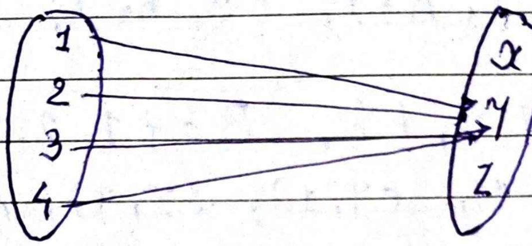
Let f be any function defined on set A.
Let $f : A \to A$ and $f(x) = x$, $\forall x \in A$,
then f is known as identity mapping & denoted by I_A.



(2) Constant function :

The function defd. from set A to set B such that $f(a) = b$, $\forall a \in A$ then f is called constant function.

Ex : $A = \{1, 2, 3, 4\}$ $B = \{x, y, z\}$

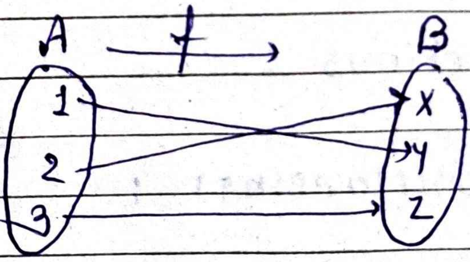


(3) One to One Mapping [Injective fn]

A function $f : A \to B$ is said to be one to one (or injective) if

$\forall x, y \in A$, $f(x) = f(y) \Longrightarrow x = y$

Ex

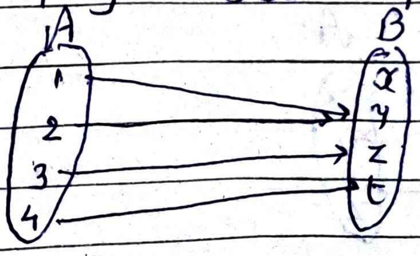


; $A = \{1, 2, 3\}$ &

$B = \{x, y, z\}$

(4) Many to one fn :

A fn $f : A \to B$ is called many to one if $x \neq y$ but $f(x) = f(y)$.

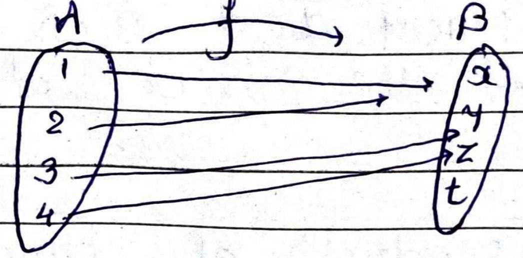


; $A = \{1, 2, 3, 4\}$

$B = \{x, y, z, t\}$

(5) Into mapping:

$f: A \to B$ is called into mapping if $\exists$ at least one element in B which is not f image of any element of A.



$A = \{1, 2, 3, 4\}$

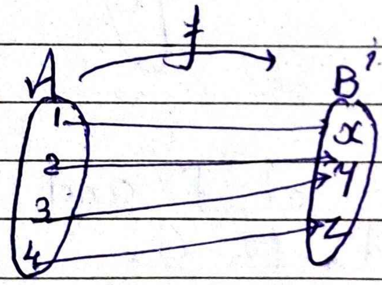
$B = \{x, y, z, t\}$

(6) Onto mapping (Surjective):

A fn $f: A \to B$ is said to be onto mapping if each element of B is the f image of at least one element of A

In this case R_f = codomain of f

i.e. $f(A) = B$.



$A = \{1, 2, 3, 4\}$, $B = \{x, y, z\}$

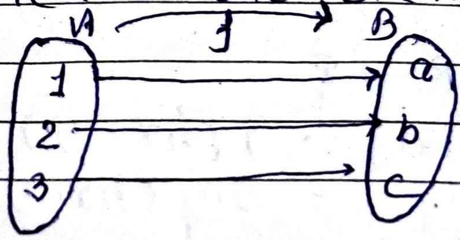
(7) One-one onto (Bijective) mapping:

A fn $f: A \to B$ which is one-one and onto both is called Bijective fn

→ Here f is one-one onto, if

(i) $f(x) = f(y) \implies x = y$ &

(ii) Every element of B is f-image of at least one element of set A.



← $A = \{1, 2, 3\}$ &

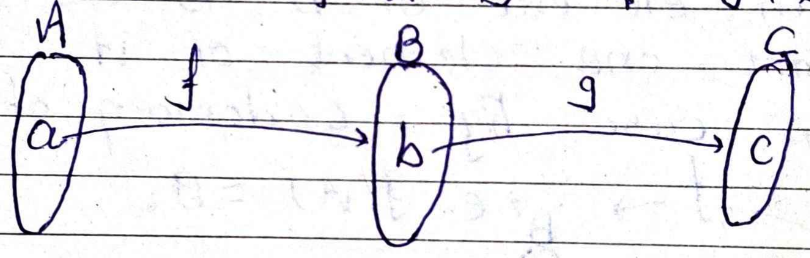
$B = \{a, b, c\}$

(8) Inverse Function :-

If $f : A \to B$ is one-one onto (Bijective) mapping, then the mapping $f^{-1} : B \to A$ which associates to element $b \in B$, to the unique element $a \in A$ such that $f(a) = b$ is called the inverse of $f : A \to B$.

(9) Composition or product of Functions :-

Let A, B, C be three sets and f be a mapping from A to B and g from B to C {i.e. $f : A \to B$ & $g : B \to C$}



Then gof, Composition of f and g is a function from A to C.

$$g \circ f(a) = g[f(a)] = g[b] = c$$

Ex : Let f, g, h be three functions from R to R as $f(x) = 4x - 5$, $g(x) = \sin x$ and $h(x) = x^2 + 1$.
Find $h \circ [g \circ f]$ and $(h \circ g) \circ f$.

Soln :- To find :- $h \circ [g \circ f]$

First $g \circ f : R \to R$

$$(g \circ f)(x) = g[f(x)] = g(4x - 5)$$
$$= \sin(4x - 5).$$

$$[h \circ [g \circ f]](x) = h[(g \circ f)(x)]$$
$$= h[\sin(4x-5)]$$
$$= [\sin^2(4x-5)] + 1$$

Similarly we can find $(h \circ g) \circ f$.

$[A; (h \circ g) \circ f = \sin^2(4x-5) + 1]$

★ Conditions for a function to be invertible

(1) If $f: A \to B$ is one-one and onto mapping then only f^{-1} exists and f^{-1} is also one-one and onto mapping.

(2) If $f: A \to B$ is one-one into, many one onto ~~and~~ then f^{-1} does not exist

★ Properties of Composite of mappings

[1] If $f: A \to B$ is one-one and onto mapping, then $f \circ f^{-1} = I_B$ & $f^{-1} \circ f = I_A$

where I_A and I_B are the identity mappings in set A and B respectively.

[2] If $f: A \to B$ and $g: B \to C$ be two one-one onto functions then $g \circ f: A \to C$ is also one-one & onto and

$$(g \circ f)^{-1} = (f^{-1} \circ g^{-1})$$

[3] Composite of functions satisfy the associative Property.
i.e. If $f: A \to B$, $g: B \to C$ & $h: C \to D$
then $(h \circ g) \circ f = h \circ (g \circ f)$

Examples based on Functions :-

[1] Let $X = \{x, y, z, k\}$ & $Y = \{1, 2, 3, 4\}$. Find which of the following are functions. Give reasons if it is not. Find range if it is a function.

(i) $f = \{(x,1), (y,2), (z,3), (k,4)\}$
(ii) $g = \{(x,1), (y,1), (k,4)\}$
(iii) $h = \{(x,1), (x,2), (x,3), (x,4)\}$
(iv) $l = \{(x,1), (y,1), (z,1), (k,1)\}$

[A : (i) Yes → $R_f = \{1,2,3,4\}$; (ii) Not a fn
(iii) Not a fn (iv) Yes, $R_f = \{1\}$.

[2] Consider f, g and h, all functions on the integers, by $f(n) = n^2$; $g(n) = n+1$ and $h(n) = n-1$. Determine (i) $h \circ f \circ g$ (ii) $g \circ f \circ h$ (iii) $f \circ g \circ h$.
[A → $n^2 + 2n$; $n^2 - 2n + 2$; n^2]

[3] Consider $f, g : R \to R$ defined by,
$f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$.
Find (i) $f \circ f$ (ii) $f \circ g$ (iii) $g \circ f$
[A → $x^4 + 6x^3 + 14x^2 + 15x + 5$; $2x^2 + 6x - 1$;
$4x^2 - 6x + 1$.]

[4] Show that the function $f: R \to R$, given by $f(x) = \cos x$, $\forall x \in R$ is neither one-one nor onto.

[5] If $f: A \to B$, $f(x) = (x-2)/(x-3)$, $A = R - \{3\}$ and $B = R - \{1\}$. show that f is bijective.

[6] Let $f: R \to R$ be defined by

$$f(x) = \begin{cases} 3x - 4, & x > 0 \\ -3x + 2, & x \leq 0. \end{cases}$$

Determine (i) $f(0)$, $f(2/3)$, $f(-2)$ &
(ii) $f^{-1}(0)$, $f^{-1}(2)$ & $f^{-1}(-7)$.

[A → 2, -6, 8 & $\{4/3\}$, $\{0, 2\}$, ϕ].

{Hint: $f^{-1}(0) = \{x \in R \mid f(x) = 0\}$

$= \{x \in R \mid x > 0 \ \& \ 3x - 4 = 0\} \cup \{x \in R \mid x \leq 0 \ \& \ -3x + 2 = 0\}$

$= \{x \in R \mid x > 0 \ \& \ x = 4/3\} \cup \{x \in R \mid x \leq 0 \ \& \ x = 2/3\}$

$= \{4/3\} \cup \phi = \{4/3\}$ }

Counting : The basics of Counting.

The Pigeonhole principle, Permutations & Combinations, Binomial Co-efficients, Generalized Permutations and Combinations, Generating Permutations & combinations.

★ Countable set :-

A set which is either empty or finite or denumerable is called Countable set.

[Note : "An infinite set A is called Denumerable if there is at least one mapping f from set A to set N such that f is one-one and onto."]

→ A set which is non-denumerable is called Uncountable set.

Ex-1 ; ϕ is countable set & $|\phi| = 0$
Ex-2 : $A = \{a, b, c, d\}$ is countable & $|A| = 4$
Ex-3 ; The set of all integers is countable

★ Properties of Countable & Uncountable Sets:

(1) A subset of a Countable set is Countable
(2) A Superset of an uncountable set is Uncountable.
(3) Union of Countable set is Countable.

(4) Cartesian Product of Countable set is Countable

(5) Union of Countable collection of Countable set is Countable.

Rmk-1 : set of real nos. in [0, 1] is an Uncountable set, so Super set of [0, 1], i.e. set of real nos. is Uncountable.

Rmk-2 :- Set of irrational no. is uncountable

Rmk-3 :- P(N) { Power set of natural nos. } is Uncountable.

Ex-1 : Prove that the set of rational no. in [0, 1] is Countable. [self ex.]

Ex-2 : P.T. the set of real nos. in [0, 1] is not Countable. [self ex.]

★ The pigeon hole Principle : [Box Argument]

The pigeon hole Principle says that if there are "many" pigeons and "a few" pigeon holes, then there must be some pigeon holes occupied by two or more pigeons.

i.e. Let A and B be finite sets and $|A| > |B|$, then for any function f from A to B, $\exists\ x, y \in A \ni f(x) = f(y)$.

Ex-1 : If 11 shoes are selected from 10 pairs of shoes then there must be a pair of matched shoes among the selection.

Theorem :- If n pegions are assigned to

m pegion holes, then one of the pigeon hole must occupy by at least

$$\left[\frac{n-1}{m}\right] + 1 \text{ pigeons.}$$

{where $\left[\frac{n-1}{m}\right]$ is the integer division of $n-1$ with m. i.e. $\left[\frac{7}{3}\right] = 2$ }

Ex-1 : If 25 people are selected then at least how many people must have been born on the same day of the week?

Soln: Here $n = 25$ and $m = 7$.
So at least $\left[\frac{n-1}{m}\right] + 1 = \left[\frac{25-1}{7}\right] + 1 = 3 + 1$
$= \boxed{4}$

at least 4 people have been born on the same day of week.

Ex-2: Show that if 5 colours are used to paint 35 scooters, then at least 7 scooters will be painted by same color.

Ex:3 Among 100 people at least how many of them were born in the same month? (GTU 2021) (Ans = 9)

Permutations and Combinations:

→ **Permutation**: "A permutation is an arrangement of a number of objects in some definite order taken some or all at a time." It is denoted by ${}^nP_r$, if n distinct objects and out of them r taken at a time.

Thm-1 The no. of different permutations of n distinct objects taken r at a time, $r \leq n$ is given by

$${}^nP_r = \frac{n!}{(n-r)!} = n(n-1)(n-2)\text{-----}(n-(r-1))$$

Thm-2: ${}^nP_n = n!$, i.e. no. of permutations of n things taken all at a time is $n!$

Ex-1: Find the value of n if.

(i) $4 \times {}^nP_3 = {}^{(n+1)}P_3$

(ii) $6 \times {}^nP_3 = 3 \times {}^{(n+1)}P_3$

(iii) $3 \times {}^nP_4 = 7 \times {}^{(n-1)}P_4$

[A: (i) 3 (ii) 5 (iii) 7]

Ex-2: How many variable names of 8 letters can be formed from the letters a, b, c, d, e, f, g, h, i if no letter is repeated. [A → $^9P_8$]

Ex-3: There are 10 persons called for an interview. "4 posts" Each one is capable to be selected for the job. How many permut. are there to select [A → $^{10}P_4$]

Ex-4 How many 6-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 if every number is to start with 30.
[A → $^7P_4$]

Ex-5 In how many ways 5 different Mathematics books and 4 different Physics books be arranged in a shelf so that all four Physics books are together?
[A → $6! \times 4! = 17280$]

Ex-6: How many Permutations can be made out of the letter of word "COMPUTER"? How many of these (i) begin with C? (ii) end with R? (iii) Begin with C and end with R? (iv) C and R occupy the end places?
[A → 40320, 5040, 5040, 720, 1440]

→ Permutations when all the objects are not distinct :-

Thm-3 The no. of permutations of n objects, of which n_1 objects are of one kind and n_2 objects are of another kind, when all are taken at a time is

$$\frac{n!}{n_1! \times n_2!}$$

Ex-1 :- Determine the no. of permutations that can be made out of the letters of the word 'PROGRAMMING'.
[A → 4989600]

Ex-2 :- There are 4 blue, 3 red and 2 black pens in a box. These are drawn one by one. Determine all the different permutations. [A → 1080] [1260]

Ex-3 :- How many different variable names can be formed by using the letters a, a, a, b, b, b, b, c, c, c ? [A → 4200]

Ex-4 :- How many 7-digit numbers can be formed using digits 1, 7, 2, 7, 6, 7, 6 ?
[A → 420]

Permutations with Repeated Objects

Thm-1 :- The no. of different permutations of n distinct objects taken r at a time when every object is allowed to repeat any number of time is given by n^r.

Ex-1: How many 4 digit numbers can be formed by using the digits 2, 4, 6, 8 when repetation of digits is allowed.
[A → 256]

Ex-2 If repetation of digits is allowed, how many 2-digits even nos. can be formed by using the digits 1, 3, 4, 6, 8?
[A → 15]

Ex-3 In how many ways can 5 software Projects be allotted to 6 final Year students when all the 5 projects are not allotted to the same student?
[A → 7776]

Circular Permutations

→ Circular permutations are the permutations of the objects placed in a circle.

Thm-1: The number of circular permutations of n different object is $(n-1)!$

Ex-1: In how many ways can the letters a, b, c, d, e, f be arranged in a circle?
(A = 120)

Ex-2 In how many ways 10 programmers can sit on a round table to discuss the project so that project leader and a particular programmer always sit together?
(A = 80640)

Ex-3: Determine the number of ways in which 5 software engineers and 6 electronics engineers can be sitted at a round table so that no two software eng. can sit together?
(A: 86400)

Combination

A Combination is a selection of some or all, objects from a set of given objects, where order of the objects does not matter." The no. of combinations of n objects, taken r at a time is represented by ^nC_r or $C(n, r)$.

Thm-1: The number of combinations of n different things, taken r at a time is given by

$$^nC_r = \frac{n!}{r!\,(n-r)!} \quad , n \geq r \geq 1$$

Rmk-1 : $^nC_n = 1$

Rmk-2 : $^nC_0 = 1$

Rmk-3 : $^nC_r = {}^nC_{n-r}$, $n \geq r \geq 1$

Ex-1: Find the value of n if,
(i) $^nC_{n-2} = 10$ & (ii) $^{20}C_{n+2} = {}^{20}C_{2n-1}$

(A → (i) 5 , (ii) 3, 8)

Ex-2: How many 16-bit strings are there containing exactly five 0's?
(A - 4368)

Ex-3: From 10 programmers in how many ways can 5 be selected when (i) A particular programmer is included every time & (ii) A part. prog. is not included at all. [A → 126 , 126]

Combination (mirrored)

Ex-4: Show that $nC_r + nC_{r-1} = {}^{n+1}C_r$, where $n \geq r \geq 1$ and n and r are natural numbers

★ Combination with Repetition :-

Thm-1 :- The number of ways to fill r slots from n categories with repetition allowed is

$$C(n+r-1, r) = C(n+r-1, n-1)$$

Ex-1 : How many ways can one fill a box holding 100 pieces of candy from 30 different types of candy?

(Hint: $n = 100$, $r = 30$ $A = 129!/(100!\,29!)$)

★ Integer solutions to Equations :-

Sometime we may be asked, how many non-negative integer solutions are there in eqn $a+b+c+d = 100$??

Here slots are 100 & categories (n) = 4 categories.

$$C(\underset{\downarrow\,r}{100} + \underset{\downarrow\,n}{4} - 1,\ 4-1) = C(100+4-1, 100)$$

$$= C(103, 100)$$

$$= \frac{103!}{100!\,3!}$$

★ Integer solutions with restrictions

Ex: How many integer soln. are there for $a+b+c+d=15$, $a \geq -3$, $b \geq 0$, $c \geq -2$ and $d \geq -1$?

→ {Hint : apply Thm-1(A)} [A: $C(24, 21)$]

Thm-1(A) The number of integer solutions to $a_1 + a_2 + a_3 + \cdots + a_n = r$, when $a_1 \geq b_1$, $a_2 \geq b_2$, ----, $a_n \geq b_n$ is

$$C(n + r - 1 - b_1 - b_2 - b_3 - \cdots - b_n,\ r - b_1 - b_2 - \cdots - b_n)$$

Thm-2: The number of ways of selecting r things from n categories with total b restrictions on the r thing is

$$C(n + r - 1 - b,\ r - b)$$

Rmk-1: The number of ways to select r things from n categories with atleast 1 thing from each category is

$C(r-1,\ r-n)$ (setting $b = n$)